ECE 332 — Homework 3 Walkthrough

Cheat sheet snippet → formula template → givens → plug → $\boxed{\text{answer}}$.

Problem 1 — 200 MHz LHC, Air → Dielectric ($\varepsilon_r = 4$) 20 pts

Setup: 200 MHz LHC plane wave, $|E| = 5$ V/m, normal incidence onto $\varepsilon_r = 4$ at $z \geq 0$. Positive max at $z = 0, t = 0$.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

Med. 1 ($z < 0$) incident side; Med. 2 ($z \geq 0$) transmitted.

	Normal ($\theta_i = 0$)	$\perp$ pol	$\parallel$ pol
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_\perp = 1 + \Gamma_\perp$	$\tau_\parallel = (1 + \Gamma_\parallel)\frac{\cos\theta_i}{\cos\theta_t}$
R refl. pwr	$\lvert\Gamma\rvert^2$	$\lvert\Gamma_\perp\rvert^2$	$\lvert\Gamma_\parallel\rvert^2$
T trans. pwr	$\lvert\tau\rvert^2 \frac{\eta_1}{\eta_2}$	$\lvert\tau_\perp\rvert^2 \frac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$	$\lvert\tau_\parallel\rvert^2 \frac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$
$R+T$ total	$T = 1 - R$	$T_\perp = 1 - R_\perp$	$T_\parallel = 1 - R_\parallel$

Notes: $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\,\sin\theta_i$; *nonmag.* $\Rightarrow \eta_2/\eta_1 = n_1/n_2$.

Intrinsic impedance η_i *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0 \xrightarrow{\mu_r = 1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega) \qquad \eta_0 = 120\pi \approx 377\,\Omega$$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction η_c $\sigma/(\omega\varepsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\varepsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\varepsilon_r}$. The j term is for $|\eta_c|, \theta_\eta$. See LOSSY MEDIA — 5 CASES for all regimes.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100\,|\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100\,|\tau|^2\,\frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

$\tilde{\mathbf{E}}$ PHASOR TEMPLATES (Air: $\varepsilon_r = \mu_r = 1$)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\varepsilon_{r,i}}}{c}\ \text{(rad/m)} \quad \Rightarrow \quad k_2 = k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

Direction table *$u \in \{x, y, z\}$ = axis perp. to boundary*

Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\tilde{\mathbf{E}}^i = a_0\hat{e}\,e^{\mp jk_1u}\ \text{(V/m)} \qquad \text{(incident wave, Med 1)}$$

$$\tilde{\mathbf{E}}^r = \Gamma\,a_0\hat{e}\,e^{\pm jk_1u}\ \text{(V/m)} \qquad \text{(reflected wave, Med 1)}$$

$$\tilde{\mathbf{E}}^t = \tau\,a_0\hat{e}\,e^{\mp jk_2u}\ \text{(V/m)} \qquad \text{(transmitted wave, Med 2)}$$

(lossy med 2: replace $e^{\mp jk_2u}$ with $e^{\mp\alpha_2u}e^{\mp j\beta_2u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x\hat{x} + a_ye^{j\delta}\hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

(a) Incident E phasor. Template: $\tilde{\mathbf{E}}^i = a_0\hat{e}\,e^{-jk_1u}$.

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y})\,e^{-j(4\pi/3)z}\ \text{V/m}$$

$$\boxed{\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y})\,e^{-j4\pi z/3}\ \text{V/m}}$$

Givens:
$f = 200$ MHz, $\omega = 2\pi f = 4\pi \times 10^8$ rad/s
$u = z$; $\mu_r = \varepsilon_{r1} = 1$ (Air)
$\hat{e} = \hat{x} + j\hat{y}$ (LHC)
$k_1 = \omega\sqrt{\varepsilon_{r1}}/c = 4\pi/3$ rad/m
$c = 3 \times 10^8$ m/s; $a_0 = 5$ V/m

(b) Reflection & transmission coefficients.

$$\eta_1(\text{Air}) = \sqrt{\tfrac{\mu_r}{\varepsilon_{r1}}}\,\eta_0 = \sqrt{1/1}(120\pi) = 120\pi\,\Omega$$

$$\eta_2(\text{med 2}) = \sqrt{\tfrac{\mu_r}{\varepsilon_{r2}}}\,\eta_0 = \sqrt{1/4}(120\pi) = 60\pi\,\Omega$$

$$\Gamma = \tfrac{\eta_2-\eta_1}{\eta_2+\eta_1} = \tfrac{60\pi-120\pi}{60\pi+120\pi} = \boxed{-\tfrac{1}{3}}\ \text{(unitless)}$$

$$\tau = 1+\Gamma = 1-\tfrac{1}{3} = \boxed{\tfrac{2}{3}}\ \text{(unitless)}$$

Givens:
$\eta_0 = 120\pi\,\Omega \approx 377\,\Omega$

(c) Reflected, transmitted, total field.

Templates: $\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e}\, e^{+jk_1 u}$, $\tilde{\mathbf{E}}^t = \tau a_0 \hat{e}\, e^{-jk_2 u}$.

$$\tilde{\mathbf{E}}^r = -\tfrac{1}{3}(5)(\hat{x}+j\hat{y})\,e^{+j(4\pi/3)z}$$

$$\boxed{\tilde{\mathbf{E}}^r = -\tfrac{5}{3}(\hat{x}+j\hat{y})\,e^{j4\pi z/3}\ \text{V/m}}$$

$$\tilde{\mathbf{E}}^t = \tfrac{2}{3}(5)(\hat{x}+j\hat{y})\,e^{-j(8\pi/3)z}$$

$$\boxed{\tilde{\mathbf{E}}^t = \tfrac{10}{3}(\hat{x}+j\hat{y})\,e^{-j8\pi z/3}\ \text{V/m}}$$

$$\tilde{\mathbf{E}}_T = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$$

$$\boxed{\tilde{\mathbf{E}}_T = 5(\hat{x}+j\hat{y})\left[e^{-j4\pi z/3} - \tfrac{1}{3}e^{j4\pi z/3}\right]\ \text{V/m}}$$

Givens:
$k_2 = \omega\sqrt{\varepsilon_{r2}}/c = 8\pi/3$
$\varepsilon_{r2} = 4$; $c = 3\times 10^8$
$\omega = 400\pi\times 10^6$

(d) % reflected and transmitted power.

$$\%P_r = 100|\Gamma|^2 = 100|-\tfrac{1}{3}|^2 = \boxed{11.1\%}$$

$$\%P_t = 100|\tau|^2\,\tfrac{\eta_1}{\eta_2} = 100|\tfrac{2}{3}|^2\,\tfrac{120\pi}{60\pi} = \boxed{88.9\%}$$

Check: $11.1\% + 88.9\% = 100\%$. *Energy conserved* ✓

Problem 2 — Same Setup, Med 2 = Poor Conductor (ε_r=2.25, σ=10^{-4}) 20 pts

Setup: Replace medium 2 with poor conductor: $\varepsilon_r = 2.25$, $\mu_r = 1$, $\sigma = 10^{-4}$ S/m. Med 1 unchanged.

TABLE 8-2 — Γ, τ, R, T (unitless) SUMMARY

Med. 1 ($z < 0$) incident side; Med. 2 ($z \geq 0$) transmitted.

	Normal ($\theta_i = 0$)	$\perp$ pol	$\parallel$ pol
Γ refl. wav	$\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\frac{\eta_2 \cos\theta_i - \eta_1 \cos\theta_t}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\frac{\eta_2 \cos\theta_t - \eta_1 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2 + \eta_1}$	$\frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_i + \eta_1 \cos\theta_t}$	$\frac{2\eta_2 \cos\theta_i}{\eta_2 \cos\theta_t + \eta_1 \cos\theta_i}$
τ rel.	$\tau = 1 + \Gamma$	$\tau_\perp = 1 + \Gamma_\perp$	$\tau_\parallel = (1 + \Gamma_\parallel)\frac{\cos\theta_i}{\cos\theta_t}$
R refl. pwr	$\lvert\Gamma\rvert^2$	$\lvert\Gamma_\perp\rvert^2$	$\lvert\Gamma_\parallel\rvert^2$
T trans. pwr	$\lvert\tau\rvert^2 \frac{\eta_1}{\eta_2}$	$\lvert\tau_\perp\rvert^2 \frac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$	$\lvert\tau_\parallel\rvert^2 \frac{\eta_1 \cos\theta_t}{\eta_2 \cos\theta_i}$
$R+T$ total	$T = 1 - R$	$T_\perp = 1 - R_\perp$	$T_\parallel = 1 - R_\parallel$

Notes: $\sin\theta_t = \sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\,\sin\theta_i$; *nonmag.* $\Rightarrow \eta_2/\eta_1 = n_1/n_2$.

Intrinsic impedance η_i *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega) \qquad \eta_0 = 120\pi \approx 377\,\Omega$$

Default $\mu_r = 1$ (air, dielectric). Use $\mu_r \neq 1$ only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction η_c $\sigma/(\omega\varepsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\varepsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\varepsilon_r}$. The j term is for $|\eta_c|, \theta_\eta$. See LOSSY MEDIA — 5 CASES for all regimes.

LOSSY MEDIA — 5 CASES

Loss tangent $\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon_0 = 8.854 \times 10^{-12}$ *F/m*

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

Type	$\sigma/\omega\varepsilon$	$\eta_c,\ \alpha,\ \beta$
Lossless ($\sigma = 0$)	0	$\eta = \eta_0/\sqrt{\varepsilon_r}$ exact $\alpha = 0$, $\beta = \omega\sqrt{\varepsilon_r}/c$
Low-loss	$< 10^{-2}$	$\eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\left(1 + j\frac{\sigma}{2\omega\varepsilon'}\right)$ $\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}$, $\beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}$
Quasi-cond.	10^{-2}–10^{2}	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$ see below
Good cond.	$> 10^2$	$\eta_c \approx (1+j)\sqrt{\pi f\mu/\sigma}$ $\alpha \approx \beta \approx \sqrt{\pi f\mu\sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\varepsilon_r}\,\eta_0$

For Γ/τ: drop j correction; use $\eta_0/\sqrt{\varepsilon_r}$. Magnetic: keep μ_r.

Low-loss quick params $\sigma/(\omega\varepsilon) \ll 1$, $\mu_r = 1$

$$\alpha \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}\ (\text{Np/m}),\quad \beta \approx \frac{\omega\sqrt{\varepsilon_r}}{c}\ (\text{rad/m}),\quad \eta_c \approx \frac{\eta_0}{\sqrt{\varepsilon_r}}\ (\Omega)$$

Exact (quasi-cond) $X = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X - 1},\quad \beta = \omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X + 1}$$

$$\theta_\eta = \tfrac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}$$

$\tilde{\mathbf{E}}$ PHASOR TEMPLATES (Air: $\varepsilon_r = \mu_r = 1$)

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega\sqrt{\varepsilon_{r,i}}}{c}\ (\text{rad/m}) \quad\Rightarrow\quad k_2 = k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

Direction table $u \in \{x, y, z\}$ = *axis perp. to boundary*

Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u \geq 0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\tilde{\mathbf{E}}^i = a_0\hat{e}\,e^{\mp jk_1u}\ (\text{V/m}) \qquad \text{(incident wave, Med 1)}$$

$$\tilde{\mathbf{E}}^r = \Gamma\,a_0\hat{e}\,e^{\pm jk_1u}\ (\text{V/m}) \qquad \text{(reflected wave, Med 1)}$$

$$\tilde{\mathbf{E}}^t = \tau\,a_0\hat{e}\,e^{\mp jk_2u}\ (\text{V/m}) \qquad \text{(transmitted wave, Med 2)}$$

(lossy med 2: replace $e^{\mp jk_2u}$ with $e^{\mp\alpha_2u}e^{\mp j\beta_2u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$.

$\hat{e}$ **by polarization** *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x\hat{x} + a_ye^{j\delta}\hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100\,|\Gamma|^2$$

% transmitted *η-ratio matters!*

$$\%P_t = 100\,|\tau|^2\,\frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.

(a) Incident E phasor. Med 1 didn't change, so same form as P1.

$$\boxed{\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y})\,e^{-j4\pi z/3}\ \text{V/m}}$$

(b) Classify medium 2, then Γ, τ.

Loss tangent:

$$\frac{\sigma}{\omega\varepsilon'} = \frac{10^{-4}}{(4\pi\times 10^8)(2.25)(8.85\times 10^{-12})} = 3.99\times 10^{-3}$$

Since $\sigma/\omega\varepsilon' \ll 10^{-2}$, **low-loss**; use $\eta_0/\sqrt{\varepsilon_r}$.

$$\eta_1 = 120\pi\,\Omega, \quad \eta_2 = \frac{120\pi}{\sqrt{2.25}} = 80\pi\,\Omega$$

$$\Gamma = \frac{80\pi - 120\pi}{80\pi + 120\pi} = \boxed{-\frac{1}{5}} \text{ (unitless)}$$

$$\tau = 1 + \Gamma = 1 - \frac{1}{5} = \boxed{\frac{4}{5}} \text{ (unitless)}$$

Givens (med 2):
$\varepsilon_{r2} = 2.25,\ \mu_{r2} = 1$
$\sigma_2 = 10^{-4}$ S/m
$\varepsilon_0 = 8.85\times 10^{-12}$ F/m
$\omega = 400\pi\times 10^6$ rad/s
$\eta_0 = 120\pi\,\Omega$

(c) Reflected, transmitted (lossy), total field.

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e}\, e^{+jk_1 z} = -\frac{1}{5}(5)(\hat{x} + j\hat{y})e^{+j4\pi z/3}$$

$$\boxed{\tilde{\mathbf{E}}^r = -(\hat{x} + j\hat{y})\, e^{j4\pi z/3} \text{ V/m}}$$

$\tilde{\mathbf{E}}^t$ replaces $e^{-jk_2 z}$ with $e^{-\alpha_2 z}e^{-j\beta_2 z}$ (lossy):

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e}\, e^{-\alpha_2 z} e^{-j\beta_2 z} = \frac{4}{5}(5)(\hat{x} + j\hat{y})e^{-0.004\pi z}e^{-j2\pi z}$$

$$\boxed{\tilde{\mathbf{E}}^t = 4(\hat{x} + j\hat{y})\, e^{-0.004\pi z}e^{-j2\pi z} \text{ V/m}}$$

$$\boxed{\tilde{\mathbf{E}}_T = 5(\hat{x} + j\hat{y})\left[e^{-j4\pi z/3} - \frac{1}{5}e^{j4\pi z/3}\right] \text{ V/m}}$$

Low-loss row (Table 5-cases):
$\alpha_2 \approx \frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}} = \frac{10^{-4}(120\pi)}{2\sqrt{2.25}} = 0.004\pi$
$= 1.26\times 10^{-2}$ Np/m
$\beta_2 \approx \frac{\omega\sqrt{\varepsilon_r}}{c} = \frac{(400\pi\times 10^6)(1.5)}{3\times 10^8} = 2\pi$ rad/m

(d) % reflected and transmitted power.

$$\%P_r = 100|\Gamma|^2 = 100|-\tfrac{1}{5}|^2 = \boxed{4\%}$$

$$\%P_t = 100|\tau|^2\,\frac{\eta_1}{\eta_2} = 100|\tfrac{4}{5}|^2\,\frac{120\pi}{80\pi} = \boxed{96\%}$$

Check: $4\% + 96\% = 100\%$. *Energy conserved* ✓

Problem 3 — Parallel-Pol, Two Layers (ε_r=6.25, 2.25) 10 pts

Setup: $\theta_i = 30°$ from air into layer 1 ($\varepsilon_r = 6.25$, 5 cm), layer 2 ($\varepsilon_r = 2.25$, 5 cm), exits to air.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin\theta_2 = \sin\theta_1\sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}} = \sin\theta_1\,\frac{n_1}{n_2}$$

Multilayer chain *stack of slabs 1 → 2 → 3 → 4*

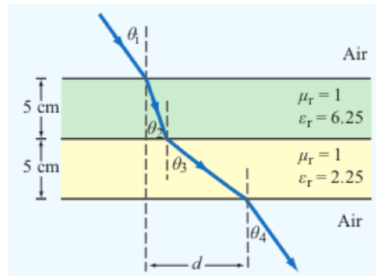
$$\sin\theta_{i+1} = \sin\theta_i\sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT



Beam offset through N slabs *thickness t_i, refracted angle inside slab i*

Slab-indexed *θ_i = angle inside slab i; t_i = slab thickness (m)*

$$d = \sum_{i=1}^{N} t_i\,\tan\theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

(a) Chained Snell's law: $\sin\theta_{i+1} = \sin\theta_i\sqrt{\varepsilon_{r,i}/\varepsilon_{r,i+1}}$

$$\sin\theta_2 = \sin 30°\sqrt{1/6.25} = 0.5(0.4) = 0.2$$

$$\boxed{\theta_2 = \sin^{-1}(0.2) = 11.54°}$$

$$\sin\theta_3 = \sin\theta_2\sqrt{6.25/2.25} = 0.2(5/3) = 1/3$$

$$\boxed{\theta_3 = \sin^{-1}(1/3) = 19.47°}$$

$$\sin\theta_4 = \sin\theta_3\sqrt{2.25/1} = (1/3)(1.5) = 0.5$$

$$\boxed{\theta_4 = 30.00° = \theta_i}$$

Givens:
$\varepsilon_{r1} = 1$ (air, $\theta_1 = 30°$)
$\varepsilon_{r2} = 6.25$ (layer 1)
$\varepsilon_{r3} = 2.25$ (layer 2)
$\varepsilon_{r4} = 1$ (air exit)

(b) Lateral distance $d = \sum_i t_i \tan\theta_{\textbf{inside slab } i}$:

$$d = t_2\tan\theta_2 + t_3\tan\theta_3$$
$$= 0.05\tan(11.54°) + 0.05\tan(19.47°)$$
$$= 0.05(0.2041) + 0.05(0.3534)$$
$$\boxed{d = 0.0279 \text{ m} = 2.79 \text{ cm}}$$

Givens:
$t_2 = t_3 = 0.05$ m (slab thickness)
$\theta_2 = 11.54°$ (inside layer 1)
$\theta_3 = 19.47°$ (inside layer 2)

Problem 4 — TE Waveguide, Find Mode, ε_r, f_c, H_y — 20 pts

Setup: TE wave, $a = 5$ cm, $b = 3$ cm, $E_x = -36\cos(40\pi x)\sin(100\pi y)\sin\big((2.4\pi\times10^{10})t - 52.9\pi z\big)$ V/m.

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0,\ H_z \neq 0$
TM mode $H_z = 0,\ E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x **form (TE)** *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

Identify TE$_{mn}$ vs TM$_{mn}$ *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. "a TE wave" → TE).
2. If $m = 0$ or $n = 0$ in the mode label ⇒ must be TE (TM forbids 0).
3. Source field $H_z(x, y)$ given ⇒ TE. Source field $E_z(x, y)$ given ⇒ TM.

TE allows at least one of $m, n \geq 1$ (other can be 0). TM needs both $m, n \geq 1$. That's why TE$_{10}$ exists but TM$_{10}$ doesn't.

ε_r FROM DISPERSION

Given ω, β, **mode** (m, n) *result is unitless*

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right] \ \text{(unitless)}$$

Inputs: ω (rad/s), β (rad/m), a, b (m), $c = 3\times10^8$ (m/s), m, n (unitless).

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$. $\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω.

TE mode *generator: $\tilde{H}_z$*

$$\tilde{H}_z = H_0 \cos\frac{m\pi x}{a} \cos\frac{n\pi y}{b}$$

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right) H_0 \cos\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$$

$$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right) H_0 \sin\frac{m\pi x}{a} \cos\frac{n\pi y}{b}$$

$$\tilde{E}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\mathrm{TE}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\mathrm{TE}}$$

$$Z_{\mathrm{TE}} = \eta/\sqrt{1-(f_c/f)^2}$$

TM mode *generator: $\tilde{E}_z$*

$$\tilde{E}_z = E_0 \sin\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$$

$$\tilde{E}_x = -\frac{j\beta}{k_c^2}\left(\frac{m\pi}{a}\right) E_0 \cos\frac{m\pi x}{a} \sin\frac{n\pi y}{b}$$

$$\tilde{E}_y = -\frac{j\beta}{k_c^2}\left(\frac{n\pi}{b}\right) E_0 \sin\frac{m\pi x}{a} \cos\frac{n\pi y}{b}$$

$$\tilde{H}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{\mathrm{TM}}, \quad \tilde{H}_y = \tilde{E}_x/Z_{\mathrm{TM}}$$

$$Z_{\mathrm{TM}} = \eta\sqrt{1-(f_c/f)^2}$$

TEM plane wave *for reference*

$$\tilde{E}_x = E_0\, e^{-j\beta z}, \quad \tilde{E}_y = \tilde{E}_z = 0$$

$$\tilde{H}_y = \tilde{E}_x/\eta, \quad \tilde{H}_x = \tilde{H}_z = 0$$

$$\eta = \sqrt{\mu/\varepsilon}\ (\Omega), \quad f_c\ \text{N/A}, \quad k = \omega\sqrt{\mu\varepsilon}$$

Properties common to TE/TM

$$f_c = \frac{u_{p0}}{2}\sqrt{(m/a)^2 + (n/b)^2}\ (\text{Hz})$$

$$\beta = k\sqrt{1-(f_c/f)^2}\ (\text{rad/m})$$

$$u_p = \frac{\omega}{\beta} = \frac{u_{p0}}{\sqrt{1-(f_c/f)^2}}\ (\text{m/s})$$

$$u_{p0} = 1/\sqrt{\mu\varepsilon}\ (\text{m/s})$$

New: Table 8-3 now contains Z_{TE}/Z_{TM}, full TEM column, and Properties Common (f_c, β, u_p, u_{p0})

— one stop for all waveguide field formulas.

(a) Mode number. Match E_x to Table 8-3 TE-row form $\cos(m\pi x/a)\sin(n\pi y/b)$.

$$\frac{m\pi}{a} = 40\pi \Rightarrow m = 40(0.05) = 2$$
$$\frac{n\pi}{b} = 100\pi \Rightarrow n = 100(0.03) = 3$$
$$\boxed{\text{Mode: TE}_{23}}$$

Givens:
$a = 0.05$ m, $b = 0.03$ m
TE wave (given), so $E_z = 0$

(b) ε_r from ε_r FROM DISPERSION box (Example 8-8).

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]$$
$$= \left(\frac{3\times10^8}{2.4\pi\times10^{10}}\right)^2\left[(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2\right]$$
$$\boxed{\varepsilon_r = 2.25}$$

Givens:
$\beta = 52.9\pi$ rad/m
$\omega = 2.4\pi\times10^{10}$ rad/s
$c = 3\times10^8$ m/s

(c) Cutoff frequency for TE_{23} from Table 8-3 Properties Common.

$$u_{p0} = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3\times10^8}{1.5} = 2\times10^8 \text{ m/s}$$
$$f_{23} = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$
$$= 10^8\sqrt{(2/0.05)^2 + (3/0.03)^2} = 10^8\sqrt{11600}$$
$$\boxed{f_c = 10.77 \text{ GHz}}$$

Givens:
$m = 2,\ n = 3$
$a = 0.05,\ b = 0.03$ m
$\varepsilon_r = 2.25$ (from b)

(d) H_y from Table 8-3 TE block: $\tilde{H}_y = \tilde{E}_x/Z_{\text{TE}}$.

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{2.25}} = 80\pi\,\Omega$$
$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1-(f_c/f)^2}} = \frac{80\pi}{\sqrt{1-(10.77/12)^2}} = 569.9\,\Omega$$
$$H_y = \frac{E_x}{Z_{\text{TE}}} = \frac{-36}{569.9}\cos(40\pi x)\sin(100\pi y)\sin(\omega t - \beta z)$$

Givens:
$f = \omega/(2\pi) = 12$ GHz
$f_c = 10.77$ GHz
$\eta_0 = 120\pi \approx 377\,\Omega$
$\varepsilon_r = 2.25$

$$\boxed{H_y = -0.0632\cos(40\pi x)\sin(100\pi y)\sin\big((2.4\pi\times10^{10})t - 52.9\pi z\big) \text{ A/m}}$$

Problem 5 — Hollow Guide Multi-Mode Travel Times 8 pts

Setup: Hollow guide ($\varepsilon_r = 1$, $u_{p0} = c$), $a = 3$ cm, $b = 2$ cm. Carrier 9.5 GHz, length $L = 100$ m. Find t for each excited mode.

CUTOFF FREQUENCY

Common cutoffs ($a > b$) *f_c formula in Table 8-3*

Mode	f_{mn}
u_{p0}	$c/\sqrt{\varepsilon_r}$ (hollow: $u_{p0} = c$)
TE_{10}	$f_{10} = u_{p0}/(2a)$ (dominant)
TE_{01}	$f_{01} = u_{p0}/(2b)$
TE_{20}	$f_{20} = u_{p0}/a = 2f_{10}$
TE_{11},TM_{11}	$f_{11} = \frac{u_{p0}}{2}\sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$. In Z_{TE}/Z_{TM} and u_g, $f_c = f_{mn}$ of the excited mode.

GROUP VELOCITY & TRAVEL TIME

Practical *$\eta_0 = 120\pi$; nonmag. media*

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}}\ (\Omega) \quad (\Leftrightarrow \sqrt{\mu/\varepsilon} \text{ in Tab. 8-3})$$

$$u_g = u_{p0}\sqrt{1 - (f_{mn}/f)^2}\ (\text{m/s})$$

$$u_p u_g = u_{p0}^2\ (\text{m}^2/\text{s}^2)$$

$t = L/u_g$ (s) travel time through guide of length L

Higher modes have lower u_g $\Rightarrow$ arrive later. (u_p, β, f_c, $Z_{TE/TM}$ in Tab. 8-3.)

Master formulas:

$$f_{mn} = \frac{u_{p0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$u_g = u_{p0}\sqrt{1 - (f_{mn}/f)^2}$$

$$t = L/u_g$$

Givens:
$f_c = 9.5\times10^9$ Hz (carrier)
$a = 0.03$ m, $b = 0.02$ m
$u_{p0} = c/\sqrt{\varepsilon_r} = 3\times10^8$ m/s
$L = 100$ m

Mode	$f_c = f_{mn}$	$u_g = u_{p0}\sqrt{1-(f_{mn}/f)^2}$	$t = L/u_g$
TE_{10}	$f_{10} = \frac{u_{p0}}{2a}$ $= \frac{3\times10^8}{2(0.03)}$ $= 5$ GHz < 9.5G **Excited**	$= 3\times10^8\sqrt{1 - \left(\frac{5G}{9.5G}\right)^2}$ $= 255.1\times10^6$ m/s	$= \frac{100}{255.1\times10^6}$ $\boxed{= 0.392\ \mu s}$
TE_{01}	$f_{01} = \frac{u_{p0}}{2b}$ $= \frac{3\times10^8}{2(0.02)}$ $= 7.5$ GHz < 9.5G **Excited**	$= 3\times10^8\sqrt{1 - \left(\frac{7.5G}{9.5G}\right)^2}$ $= 184.1\times10^6$ m/s	$= \frac{100}{184.1\times10^6}$ $\boxed{= 543.1\ \text{ns}}$
TE_{20}	$f_{20} = \frac{u_{p0}}{a}$ $= \frac{3\times10^8}{0.03}$ $= 10$ GHz > 9.5G **Not excited**	N/A	N/A
TE_{11} TM_{11}	$f_{11} = \frac{u_{p0}}{2}\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}$ $= \frac{3\times10^8}{2}\sqrt{\frac{1}{0.03^2} + \frac{1}{0.02^2}}$ $= 9.014$ GHz < 9.5G **Excited**	$= u_{p0}\sqrt{1 - \left(\frac{9.014}{9.5}\right)^2}$ $= 95.10\times10^6$ m/s	$= \frac{100}{95.10\times10^6}$ $\boxed{= 1.05\ \mu s}$

Higher modes arrive later: as $f_{mn} \to f$, $u_g \to 0$, so closer-to-cutoff modes have longer travel time. Group delay spread = max t - min $t \approx 1.05 - 0.392 \approx 0.66\ \mu s$ across all excited modes.